Classical and Bayesian Estimation for Block Progressively Censored Competing Risks from the Kumaraswamy Distribution

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Abstract

We investigate failure times when an item can fail from either of two causes and some surviving items are removed during testing. We model the failure times with a Kumaraswamy distribution under block progressive Type-II censoring. The shape parameter is shared across blocks, while the cause parameters can vary. We develop maximum likelihood and Bayesian estimators, both with and without an order restriction within each block. Classical inference uses a profile likelihood; Bayesian inference uses either an independent scale prior or a hierarchical Gamma prior. We explain uncertainty intervals, posterior checks, and reliability quantities, and provide a reproducible simulation design and a real-data application. The exploratory simulation comprises 60 saved datasets and compares all six methods with explicit diagnostic screening. All six methods have 60 diagnostically acceptable fits, with comparisons paired on identical original dataset IDs. Hierarchical unrestricted Bayes has lower observed MSE than independent unrestricted Bayes for all primary parameters in this small design. Incorrect ordering can severely reduce cause-specific interval coverage. A published 39-failure appliance dataset illustrates the effect of imposing an ordering assumption that the recorded cause frequencies do not support.
Keywords: Kumaraswamy distribution; competing risks; block progressive censoring; maximum likelihood estimation; Bayesian estimation; order restriction; credible interval; simulation.

1 Introduction

In reliability studies, an item can fail for different reasons. We observe the time of the first failure and, when known, its cause. Sometimes testing ends before every item fails. Progressive Type-II censoring allows a predetermined number of surviving items to be removed after each observed failure. In block progressive censoring, this procedure is conducted separately in different groups or facilities.

The Kumaraswamy distribution is a flexible model for times measured on a fixed interval $(0, 1)$. Previous studies consider block progressive competing risks under another distribution [2], progressively censored Kumaraswamy competing risks [3], and Kumaraswamy competing risks with a different hybrid censoring scheme [4]. We combine the specified block censoring design with a shared-shape Kumaraswamy model. The common shape, independence assumptions and proposed order restriction are modeling choices; they require justification for a particular application.

The aim of this study is to see how the estimation methods behave when the two causes have different strengths, when their parameters are equal, and when an assumed order is incorrect. We first describe the data and likelihood, then derive the classical and Bayesian approaches. Finally, we explain the simulation design, what each performance measure tells us, and how we apply the model to observed appliance-failure data.

2 Model and Likelihood

2.1 The Kumaraswamy competing-risks model

Suppose there are k independent blocks. In block i, each item has two independent latent failure times

$$T_{iu1} \sim \mathrm{Kw}(a, b_{i1}), \qquad T_{iu2} \sim \mathrm{Kw}(a, b_{i2}), \qquad a, b_{i1}, b_{i2} > 0. \tag{1}$$

The shape parameter a is common to all blocks and causes. The parameters b_{i1}, b_{i2} may differ between blocks. Items are independent, and items within the same block have identical distributions. We observe $X_{iu} = \min(T_{iu1}, T_{iu2})$ and the cause $\delta_{iu} \in \{1, 2\}$ of that first failure. All recorded causes are assumed known.

For $T \sim \mathrm{Kw}(a, b)$ and $0 < t < 1$,

$$F(t) = 1 - (1 - t^a)^b, \qquad f(t) = ab\, t^{a-1}(1 - t^a)^{b-1}, \tag{2}$$

$$S(t) = (1 - t^a)^b, \qquad h(t) = \frac{ab\, t^{a-1}}{1 - t^a}. \tag{3}$$

Put $B_i = b_{i1} + b_{i2}$. Independence gives

$$S_i(t) = P(X_{iu} > t) = (1 - t^a)^{B_i}, \qquad X_{iu} \sim \mathrm{Kw}(a, B_i). \tag{4}$$

The joint density of the first failure time and cause c is

$$g_{ic}(t) = f(t; a, b_{ic}) S(t; a, b_{i,3-c}) = ab_{ic} t^{a-1}(1 - t^a)^{B_i - 1}. \tag{5}$$

Its integral is the probability of cause c:

$$P(\delta_{iu} = c) = \frac{b_{ic}}{B_i}. \tag{6}$$

Since $g_{ic}(t)$ is the minimum-lifetime density multiplied by this constant probability, cause and observed lifetime are independent under our model.

2.2 Block progressive censoring and likelihood

In block i, begin with n_i items. Observe m_i failures at ordered times $0 < x_{i1} < \cdots < x_{im_i} < 1$, recording each cause δ_{ij}. Immediately after failure j, randomly withdraw r_{ij} surviving items, independently of their future lifetimes and causes. The fixed withdrawal scheme satisfies

$$n_i = m_i + \sum_{j=1}^{m_i} r_{ij}. \tag{7}$$

The number at risk immediately before failure j is

$$Y_{ij} = n_i - (j - 1) - \sum_{\ell < j} r_{i\ell}. \tag{8}$$

An observed failure contributes the cause-specific density $g_{i,\delta_{ij}}(x_{ij})$. Each withdrawn survivor contributes $S_i(x_{ij})$. For the ordered failure times and recorded cause sequence, the block likelihood is

$$L_i = C_i \prod_{j=1}^{m_i} g_{i,\delta_{ij}}(x_{ij}) S_i(x_{ij})^{r_{ij}}, \qquad C_i = \prod_{j=1}^{m_i} Y_{ij}. \tag{9}$$

The constant C_i accounts for the possible items that could fail at each stage; it does not depend on the unknown parameters. The blocks are independent, so $L = \prod_i L_i$.

Let d_{ic} be the number of observed failures from cause c in block i, $M = \sum_i m_i$, and define

$$T = \sum_{i,j} \log x_{ij}, \qquad H_i(a) = \sum_j \log(1 - x_{ij}^a), \tag{10}$$

$$W_i(a) = -\sum_j (r_{ij} + 1) \log(1 - x_{ij}^a), \qquad C = \prod_i C_i. \tag{11}$$

Substituting (4) and (5) into (9), then taking logarithms, gives the full log-likelihood

$$\boxed{\ell = \log C + M \log a + (a-1)T + \sum_{i,c} d_{ic} \log b_{ic} - \sum_i B_i W_i(a) - \sum_i H_i(a).} \tag{12}$$

Terms with a zero count are interpreted through the continuous convention $0 \log b = 0$.

3 Maximum Likelihood Estimation

3.1 Unrestricted maximum likelihood estimation

To maximize (12), differentiate with respect to each b_{ic}:

$$\frac{\partial \ell}{\partial b_{ic}} = \frac{d_{ic}}{b_{ic}} - W_i(a). \tag{13}$$

When $d_{ic} > 0$, setting this score to zero yields

$$\boxed{\widehat{b}_{ic}(a) = \frac{d_{ic}}{W_i(a)}, \qquad \widehat{B}_i(a) = \frac{m_i}{W_i(a)}.} \tag{14}$$

If $d_{ic} = 0$, the supremum occurs as $b_{ic} \downarrow 0$: this is a boundary limit, not an attained maximum under the strict positivity assumption.

Substitute (14) into (12). The resulting profile log-likelihood, up to terms constant in a, is

$$\ell_p(a) = M \log a + (a-1)T - \sum_i m_i \log W_i(a) - \sum_i H_i(a) + \text{constant}. \tag{15}$$

Its score is

$$U_p(a) = \frac{M}{a} + T - \sum_i m_i \frac{W_i'(a)}{W_i(a)} - \sum_i H_i'(a). \tag{16}$$

We solve $U_p(a) = 0$ numerically with a bracketed root method, and substitute $\widehat{a}$ into (14). The profile is strictly concave in a and has a unique finite maximizer if at least one block has two or more distinct observed failure times. If every block has only one observed failure, the profile likelihood is unbounded as $a \to \infty$. A finite positive-parameter maximum also requires both cause counts to be positive in each block. These conclusions concern the mathematical likelihood; numerical failure of an optimizer is a separate issue.

3.2 Maximum likelihood estimation with an order restriction

Assume, for each block separately,

$$b_{i1} \le b_{i2}. \tag{17}$$

This is an additional modeling assumption, not an order imposed between blocks. Set

$$B_i = b_{i1} + b_{i2}, \qquad p_i = \frac{b_{i1}}{B_i}. \tag{18}$$

The restriction is simply $p_i \le 1/2$, and the log-likelihood separates into a part involving (a, B_i) and a cause-count part

$$\ell = \ell_{\text{time}}(a, \mathbf{B}) + \sum_i \{d_{i1} \log p_i + d_{i2} \log(1 - p_i)\}. \tag{19}$$

The cause-count part is maximized at d_{i1}/m_i without a restriction. Under (17), its maximum over $0 \le p_i \le 1/2$ is therefore

$$\boxed{\widetilde{p}_i = \min\left(\frac{d_{i1}}{m_i}, \frac{1}{2}\right), \qquad \widetilde{b}_{i1} = \widehat{B}_i \widetilde{p}_i, \quad \widetilde{b}_{i2} = \widehat{B}_i(1 - \widetilde{p}_i).} \tag{20}$$

Thus we keep the unrestricted pair if it satisfies the order; otherwise both parameters are pooled at $\widehat{B}_i/2$. The optimized cause-count term depends only on counts, not on a or B_i. Consequently,

$$\boxed{\widetilde{a} = \widehat{a}, \qquad \widetilde{B}_i = \widehat{B}_i.} \tag{21}$$

In particular, the restricted and unrestricted fitted block survival and hazard functions are identical. A strict violation in observed cause proportions causes a loss of maximized log-likelihood under the restriction. If $d_{i1} = 0$, the restricted solution also lies on a zero boundary; if $d_{i2} = 0$, pooling yields two positive equal parameters. The equality constraint is permitted, but a zero parameter is not in the original strictly positive space.

3.3 Observed information and confidence intervals

The observed information is the negative Hessian of the log-likelihood. To simplify its expression, use the parameter vector $q = (a, B_1, \ldots, B_k, p_1, \ldots, p_k)$. Put $w_{ij} = r_{ij} + 1$, $u_{ij} = -\log x_{ij}$ and

$$P_{ij} = \frac{u_{ij}}{e^{a u_{ij}} - 1}, \qquad Q_{ij} = \frac{u_{ij}^2 e^{-a u_{ij}}}{(1 - e^{-a u_{ij}})^2}. \tag{22}$$

Differentiating (19), the nonzero entries of $J(q) = -\nabla^2 \ell(q)$ are

$$J_{aa} = \frac{M}{a^2} + \sum_{i,j} (B_i w_{ij} - 1) Q_{ij}, \qquad J_{aB_i} = -\sum_j w_{ij} P_{ij}, \tag{23}$$

$$J_{B_i B_i} = \frac{m_i}{B_i^2}, \qquad J_{p_i p_i} = \frac{d_{i1}}{p_i^2} + \frac{d_{i2}}{(1 - p_i)^2}. \tag{24}$$

All other entries vanish, except the symmetric counterparts. For a regular unrestricted interior MLE, $J(\widehat{q})^{-1}$ estimates its covariance matrix, and the square roots of its diagonal entries give standard errors. For a positive parameter θ, an approximate 95% log-scale interval is

$$\left[\widehat{\theta} \exp\left(-1.96 \frac{\mathrm{SE}(\widehat{\theta})}{\widehat{\theta}}\right), \ \widehat{\theta} \exp\left(1.96 \frac{\mathrm{SE}(\widehat{\theta})}{\widehat{\theta}}\right)\right]. \tag{25}$$

The covariance for the original b parameters follows from the Jacobian of $b_{i1} = B_i p_i$ and $b_{i2} = B_i(1 - p_i)$. These intervals are large-sample approximations, not exact guarantees.

Under our model the observed cause count satisfies $d_{i1} \sim \text{Binomial}(m_i, p_i)$, conditional on the fixed number of failures. Thus an exact Clopper–Pearson confidence interval can be formed for each p_i. Under the order restriction, intersect this interval with $[0, 1/2]$. If the intersection is empty, report it as empty; do not force it to contain the restricted estimate. Ordinary unrestricted Wald intervals are not automatically valid for cause parameters at active constraints. The currently developed methods do not assign such intervals to restricted individual b_{ic} parameters.

3.4 Reliability estimation

Using $\widehat{a}$ and $\widehat{B}_i$, the fitted block survival, hazard and median time to failure are

$$\widehat{S}_i(t) = (1 - t^{\widehat{a}})^{\widehat{B}_i}, \tag{26}$$

$$\widehat{h}_i(t) = \frac{\widehat{a}\widehat{B}_i t^{\widehat{a}-1}}{1 - t^{\widehat{a}}}, \tag{27}$$

$$\widehat{v}_i = \left(1 - 2^{-1/\widehat{B}_i}\right)^{1/\widehat{a}}. \tag{28}$$

The fitted cause-c cumulative incidence is

$$\widehat{G}_{ic}(t) = \widehat{p}_{ic}\{1 - \widehat{S}_i(t)\}, \qquad \widehat{p}_{ic} = \frac{\widehat{b}_{ic}}{\widehat{B}_i}. \tag{29}$$

For regular estimates, approximate standard errors of these functions can be obtained by the delta method: if $g(q)$ is a reliability quantity, $\widehat{\mathrm{Var}}\{g(\widehat{q})\} = \nabla g(\widehat{q})^{\mathsf{T}} J(\widehat{q})^{-1} \nabla g(\widehat{q})$. Overall survival, hazard and median use only (a, B_i) and are unchanged by the within-block restriction. Cause-specific incidence also depends on p_i and may change.

4 Bayesian Estimation

We use the same likelihood in (12). Throughout, Gamma distributions are parameterized by *shape* and *rate*. The squared-error Bayes estimate is a posterior mean, provided that mean exists. We study two prior structures, each with and without the within-block order restriction (17). The notation $\pi(\cdot \mid D)$ denotes a posterior density, up to proportionality where normalization is not written.

4.1 Independent scale prior without ordering

Consider the improper independent scale prior

$$\pi_I(a, \mathbf{b}) \propto a^{-1} \prod_{i=1}^{k} \prod_{c=1}^{2} b_{ic}^{-1}, \qquad a, b_{ic} > 0. \tag{30}$$

This is a scale-invariant prior specification, *not* a claim that it is the joint Jeffreys prior. Combining (30) with (12), the posterior kernel is

$$\pi_I(a, \mathbf{b} \mid D) \propto a^{M-1} e^{(a-1)T - \sum_i H_i(a)} \prod_{i,c} b_{ic}^{d_{ic}-1} e^{-b_{ic} W_i(a)}. \tag{31}$$

For $d_{ic} > 0$, the full conditionals are independent Gamma distributions:

$$b_{ic} \mid a, D \sim \mathrm{Gamma}(d_{ic}, W_i(a)). \tag{32}$$

Integrating out all cause parameters gives a one-dimensional marginal kernel for a:

$$\boxed{h_I(a) \propto a^{M-1} \exp\left\{(a-1)T - \sum_i H_i(a)\right\} \prod_i W_i(a)^{-m_i}.} \tag{33}$$

This density can be sampled using a log-scale slice update for $\eta = \log a$, including the transformation Jacobian e^{η}. The b_{ic} can then be drawn conditionally from (32).

The improper prior requires explicit posterior propriety checks. Set $y_{ij} = -\log x_{ij}$, $y_i^* = \min_j y_{ij}$, and $D_* = \sum_{i,j}(y_{ij} - y_i^*)$. For the unrestricted independent-scale analysis, the posterior is proper when every $d_{ic} > 0$, $M > 1$ and $D_* > 0$; outside that domain the implementation does not return posterior estimates. Moreover,

$$E_I(b_{ic}^r \mid D) < \infty \quad \Longleftrightarrow \quad D_* > r y_i^*, \qquad r > 0. \tag{34}$$

A proper posterior does not by itself guarantee a finite posterior mean or variance. The implementation checks moment existence separately rather than relying on finite sampled averages.

4.2 Independent scale prior with within-block ordering

For each block use $B_i = b_{i1} + b_{i2}$ and $p_i = b_{i1}/B_i$. The Jacobian is B_i. Transforming the joint posterior in (31) gives the joint kernel

$$\pi_I(a, \mathbf{B}, \mathbf{p} \mid D) \propto a^{M-1} e^{(a-1)T - \sum_i H_i(a)} \prod_i \left\{ B_i^{m_i - 1} e^{-B_i W_i(a)} p_i^{d_{i1}-1} (1-p_i)^{d_{i2}-1} \right\}. \tag{35}$$

Integrating over B_i and p_i recovers the marginal density $h_I(a)$ in (33). In particular,

$$B_i \mid a, D \sim \text{Gamma}(m_i, W_i(a)), \qquad p_i \mid D \sim \text{Beta}(d_{i1}, d_{i2}). \tag{36}$$

The restriction $b_{i1} \le b_{i2}$ is equivalent to $p_i \le 1/2$. Under the restricted posterior, only p_i is truncated to $(0, 1/2]$; $h_I(a)$ and the conditional distribution of each B_i are unchanged. Thus the full posterior distributions of overall block survival, hazard and median are invariant to ordering *for this independent prior*. Cause proportions and cause-specific incidence need not be invariant. Restricted posterior means are calculated from constrained draws, rather than by projecting or pooling unrestricted posterior means.

The implementation conservatively requires both cause counts positive even under restriction. This is a computational support rule, not a mathematical assertion that all restricted zero-count posteriors are improper: some cases with $d_{i2} = 0 < d_{i1}$ can be proper after conditioning on $p_i \le 1/2$.

4.3 Hierarchical Gamma prior without ordering

We also consider a proper hierarchical model in which cause-specific parameters share information across blocks:

$$a \sim \text{Gamma}(A_a, B_a), \tag{37}$$

$$b_{ic} \mid \kappa_c, \lambda_c \overset{\text{ind}}{\sim} \text{Gamma}(\kappa_c, \lambda_c), \tag{38}$$

$$\kappa_c \sim \text{Gamma}(A_c, B_c), \qquad \lambda_c \sim \text{Gamma}(C_c, D_c), \qquad c = 1, 2. \tag{39}$$

All fixed hyperparameters are positive and the two causes have separate hyperparameter pairs. Up to a factor independent of unknowns, the joint posterior is

$$\pi_H(a, \mathbf{b}, \boldsymbol{\kappa}, \boldsymbol{\lambda} \mid D) \propto a^{M + A_a - 1} e^{-B_a a + (a-1)T - \sum_i H_i(a)} \times \prod_{c=1}^{2} \left[\frac{\kappa_c^{A_c - 1} e^{-B_c \kappa_c} \lambda_c^{k\kappa_c + C_c - 1} e^{-D_c \lambda_c}}{\Gamma(\kappa_c)^k} \prod_i b_{ic}^{\kappa_c + d_{ic} - 1} e^{-(\lambda_c + W_i(a)) b_{ic}} \right]. \tag{40}$$

In particular, neither $\lambda_c^{k\kappa_c}$ nor $\Gamma(\kappa_c)^{-k}$ may be omitted: both depend on unknown hyperparameters. The conjugate full conditionals are

$$b_{ic} \mid \text{rest}, D \sim \text{Gamma}(\kappa_c + d_{ic}, \lambda_c + W_i(a)), \tag{41}$$

$$\lambda_c \mid \text{rest}, D \sim \text{Gamma}\left(k\kappa_c + C_c, D_c + \sum_i b_{ic}\right). \tag{42}$$

The nonconjugate conditionals for a and κ_c are updated by slice sampling on the log scale. For reference,

$$\log \pi(\kappa_c \mid \text{rest}, D) = (A_c - 1)\log \kappa_c - B_c\kappa_c - k \log \Gamma(\kappa_c) + \kappa_c \left(k \log \lambda_c + \sum_i \log b_{ic}\right) + \text{const}, \tag{43}$$

$$\log \pi(a \mid \text{rest}, D) = (M + A_a - 1)\log a - B_a a + (a-1)T - \sum_i H_i(a) - \sum_i B_i W_i(a) + \text{const}. \tag{44}$$

The change of variables in each log-scale slice update includes its Jacobian. For valid progressively censored data, positive fixed Gamma hyperparameters lead to a proper posterior. The code distinguishes verified finite moments from moments that are not established by its sufficient bounds.

4.4 Hierarchical Gamma prior under within-block ordering

The restricted hierarchical model conditions the *whole joint posterior* in (40) on the event

$$\mathcal{C} = \bigcap_{i=1}^{k} \{b_{i1} \le b_{i2}\}. \tag{45}$$

The Gibbs update for b_{i1} is its Gamma conditional truncated above at the current b_{i2}; the update for b_{i2} is its Gamma conditional truncated below at the newly sampled b_{i1}. The remaining full conditionals retain their forms because the event indicator is then fixed. We do not add separate hyperparameter-dependent normalizing factors corresponding to an independently truncated conditional prior; doing so would define a different model.

Unlike the independent scale-prior construction, the restriction here can alter the marginal posterior of a, the totals B_i and hence overall reliability. No across-block ordering is imposed.

4.5 Bayesian estimates, intervals, diagnostics and limitations

All four Bayesian analyses use posterior means under squared-error loss where the requisite first moments are established. For any fixed $0 < t < 1$, the posterior reliability functions are calculated for each joint draw: $S_i(t)$, $h_i(t)$, the median v_i, the cause probabilities p_i and the cause-specific cumulative incidence $G_{ic}(t)$. Averaging draw-wise functions is generally different from evaluating a nonlinear function at posterior mean parameters.

Marginal intervals are the shortest contiguous intervals containing at least a prespecified fraction (typically 0.95) of pooled retained draws. These approximate highest-posterior-density (HPD) intervals for suitable unimodal marginals; they are not guaranteed to represent disconnected HPD regions and are not invariant under reparameterization. Credible probabilities are posterior probabilities, not empirical frequentist coverage claims.

Sampling uses multiple independently initialized chains, a warm-up phase, and retained draws. Diagnostics include rank-normalized and folded split $\widehat{R}$, bulk and tail effective sample sizes, and

Monte Carlo standard errors when posterior second moments are established. The configured default checks are $\widehat{R} \leq 1.01$ and bulk and tail ESS ≥ 400; these are diagnostics rather than proofs of convergence. Numerical failures, convergence failures, posterior impropriety and unsupported moment conditions are recorded explicitly. The sensitivity of hierarchical estimates to prior choices remains a limitation. In the numerical study below, all reported Bayesian point estimates and intervals are restricted to fits that passed the recorded diagnostic and moment checks.

5 Numerical Illustration

5.1 Simulation Study

The simulation uses $k = 2$ independent blocks and two latent failure causes per item. For each unit independently draw $U_{iuc} \sim \text{Uniform}(0, 1)$ and set

$$T_{iuc} = \left[1 - (1 - U_{iuc})^{1/b_{ic}}\right]^{1/a}, \qquad X_{iu} = \min(T_{iu1}, T_{iu2}), \quad \delta_{iu} = \arg\min_{c=1,2} T_{iuc}. \tag{46}$$

At every observed failure the item with the earliest remaining lifetime fails, after which the prescribed number r_{ij} of surviving items is withdrawn uniformly at random. Merely keeping the first m_i complete-data order statistics would not implement progressive censoring correctly. Every observed dataset is saved with scenario, seed, times, causes, withdrawals and true parameters; all six estimation procedures receive that same observed dataset.

The submission-scale experiment uses $a = 1.5$, $n_1 = n_2 = 50$, $m_1 = m_2 = 30$, and $r_{i,m_i} = 20$ with all earlier removals zero. Three principal scenarios are:

True ordering	(b_{11}, b_{12})	(b_{21}, b_{22})
Strict	$(0.5, 1.5)$	$(0.75, 2.25)$
Equality	$(1, 1)$	$(1.5, 1.5)$
Violated	$(1.5, 0.5)$	$(2.25, 0.75)$

All scenarios preserve $B_1 = 2$ and $B_2 = 3$, isolating cause allocation while keeping the marginal minimum-lifetime distributions fixed. There are 20 generated datasets in each scenario. This makes the study useful for illustrating patterns, but not for drawing high-precision conclusions about coverage or differences in MSE. Other sample sizes and censoring plans would be useful in a larger study.

For each scenario, method m and parameter θ, let $\mathcal{V}_{m\theta}$ denote the repetitions with a valid reported estimate and $N_{m\theta} = |\mathcal{V}_{m\theta}|$. We compute

$$\overline{\widehat{\theta}}_m = \frac{1}{N_{m\theta}} \sum_{r \in \mathcal{V}_{m\theta}} \widehat{\theta}_m^{(r)}, \tag{47}$$

$$\text{AB}_m(\theta) = \left|\overline{\widehat{\theta}}_m - \theta_0\right|, \qquad \text{MSE}_m(\theta) = \frac{1}{N_{m\theta}} \sum_{r \in \mathcal{V}_{m\theta}} \left(\widehat{\theta}_m^{(r)} - \theta_0\right)^2.$$

For a supported interval $[L_m^{(r)}, U_m^{(r)}]$, use the valid interval subset $\mathcal{I}_{m\theta}$ with size $N_{m\theta}^I$ and calculate

$$\text{CP}_m(\theta) = \frac{1}{N_{m\theta}^I} \sum_{r \in \mathcal{I}_{m\theta}} \mathbf{1}\{L_m^{(r)} \leq \theta_0 \leq U_m^{(r)}\}, \qquad \text{AW}_m(\theta) = \frac{1}{N_{m\theta}^I} \sum_{r \in \mathcal{I}_{m\theta}} (U_m^{(r)} - L_m^{(r)}). \tag{48}$$

Every summary reports its valid numerator and denominator as well as method-specific numerical and MCMC failure rates. Empty confidence sets, where defined, have zero coverage and zero set length; unavailable individual constrained-parameter intervals are marked not applicable, never fabricated. Restricted-model results under violated true ordering are a deliberate misspecification assessment. A method's smaller conditional MSE should not conceal a poor success rate.

How to read the numerical results

Absolute bias (AB) measures systematic departure from the true parameter: smaller values are preferable. Mean squared error (MSE) also penalizes variability, so a method can have small bias but still have a large MSE. Coverage probability (CP) checks how often an interval contains the true parameter; we compare it with the nominal 95% target. Average width (AW) measures interval precision, but a narrow interval is only attractive when its coverage is adequate. Because the simulation is small and Bayesian fits may not complete for every dataset, we give the number of usable replications beside every metric. For fair head-to-head MSE comparisons, we use the same successfully analyzed datasets for both methods.

Completed local execution and validation

All 60 original datasets were retained with their recorded identities and seeds. The final registry has 360 valid fits out of 360 planned combinations, with 0 missing and 0 unresolved invalid combinations. All computation was executed locally on the MacBook (10 CPU cores, 16 GiB RAM), with three worker processes and one BLAS/OpenMP thread per worker.

The baseline MCMC uses four independent chains, 1,000 warm-up and 2,000 retained iterations per chain. Failed diagnostics trigger a predetermined increase to 2,000/8,000 warm-up/retained iterations, then 4,000/32,000 if needed. The original method–dataset seed, priors, likelihood and thresholds ($\widehat{R} \leq 1.01$, bulk and tail ESS ≥ 400 for every monitored parameter and reliability quantity) are retained. Historical successes are reused without refitting. Historical failures remain available. The first acceptable effort level is selected; no favorable-seed selection is performed.

The historical counts were 60, 60, 31, 31, 31 and 27. There were 109 uncompleted combinations and 11 restricted hierarchical diagnostic failures. The completion computed 120 successful replacements/additions and preserved eight additional baseline diagnostic failures before longer retries. All 19 historical/new failed records are retained. All additional successful fits passed by the second effort level; the third level was unnecessary.

The hierarchical shape update now evaluates the identical conditional kernel $(M + A_a - 1)\log a + (a-1)T - B_a a - \sum_i H_i(a) - \sum_i B_i W_i(a)$ from validated block arrays rather than rebuilding each observation contribution at every slice proposal. Independent row-likelihood ratio regression checks verify this optimization. The posterior target and Gibbs/slice kernels are unchanged.

Table 1: Verified valid fits. Every method uses the same original 20 datasets within each scenario.

Method	Strict	Equality	Violation	Total
MLE U	20	20	20	60
MLE R	20	20	20	60
Independent U	20	20	20	60
Independent R	20	20	20	60
Hierarchical U	20	20	20	60
Hierarchical R	20	20	20	60

Fair six-method parameter accuracy

The six-way comparison takes the intersection of actual original dataset IDs and verifies their data hashes and truths. It contains all 20 datasets in every scenario for all five primary parameters. Tables below report AB and MSE for both causes in both blocks; no ranking is based on incompatible subsets. Full precision estimates, auxiliary quantities, 15 pairwise contrasts, paired squared-error-difference Monte Carlo SEs, and dataset memberships are saved in the versioned CSV exports.

Table 2: Strict ordering: absolute bias on the identical 20 datasets. All denominators are 20.

Method	a	b_{A1}	b_{A2}	b_{B1}	b_{B2}
MLE U	0.099645	0.067722	0.212042	0.150948	0.555558
MLE R	0.099645	0.067722	0.212042	0.150948	0.555558
Independent U	0.090574	0.072589	0.229392	0.169395	0.615072
Independent R	0.090213	0.065758	0.236511	0.158939	0.619236
Hierarchical U	0.071780	0.100886	0.203799	0.152663	0.440865
Hierarchical R	0.075417	0.095180	0.220013	0.148532	0.464578

Table 3: Strict ordering: mean squared error on the identical 20 datasets. All denominators are 20.

Method	a	b_{A1}	b_{A2}	b_{B1}	b_{B2}
MLE U	0.051452	0.045749	0.369954	0.128948	0.949672
MLE R	0.051452	0.045749	0.369954	0.128948	0.949672
Independent U	0.049317	0.047925	0.393239	0.140906	1.071643
Independent R	0.049215	0.043173	0.389503	0.135021	1.062474
Hierarchical U	0.036386	0.045906	0.314938	0.105174	0.642633
Hierarchical R	0.036750	0.043125	0.318495	0.103559	0.652734

Strict ordering. Neither block activates classical pooling, so both MLEs coincide. Hierarchical unrestricted shape MSE is 0.036386, versus 0.051452 for MLE (29.3% lower in this experiment). The hierarchical prior also lowers the observed MSE of both cause coefficients in block B. The independent Bayesian procedure does not uniformly improve cause-parameter MSE over MLE. Restriction changes Bayesian cause allocation slightly, with mixed effects across coefficients.

Table 4: Equality: absolute bias on the identical 20 datasets. All denominators are 20.

Method	a	b_{A1}	b_{A2}	b_{B1}	b_{B2}
MLE U	0.004033	0.014241	0.056455	0.061062	0.067459
MLE R	0.004033	0.081653	0.123867	0.028251	0.156772
Independent U	0.004822	0.005685	0.066396	0.088925	0.093642
Independent R	0.005117	0.164463	0.223269	0.160703	0.341958
Hierarchical U	0.012988	0.011302	0.078076	0.028199	0.043036
Hierarchical R	0.011571	0.145226	0.231858	0.200870	0.278959

Equality. Restricted MLE reduces block-A cause-1 MSE from 0.126493 to 0.096261, while cause-2 MSE rises from 0.182972 to 0.190394. The corresponding Bayesian restrictions also improve cause 1 and worsen cause 2 in both blocks. Pooling and one-sided posterior truncation therefore trade variance reduction against asymmetric estimation bias even at a correct equality boundary.

Ordering violation. Restricted MLE changes block-A cause-1 MSE from 1.000758 to 0.396180, but cause-2 MSE rises from 0.064089 to 0.829559. Both Bayesian families show the same direction of tradeoff, including block B. A lower MSE for the larger true coefficient does not validate the false restriction; bias and variance can offset for one coordinate while worsening the other. The complete data-ID comparison resolves the earlier one-dataset restricted-hierarchical subset limitation.

Interval coverage and precision

Classical entries below are positive log-Wald confidence intervals; Bayesian entries are 95% shortest contiguous posterior-draw credible intervals. These are distinct procedures. Restricted individual-cause classical intervals remain unsupported and are not replaced by ordinary Wald intervals. All

Table 5: Equality: mean squared error on the identical 20 datasets. All denominators are 20.

Method	a	b_{A1}	b_{A2}	b_{B1}	b_{B2}
MLE U	0.037730	0.126493	0.182972	0.371354	0.174918
MLE R	0.037730	0.096261	0.190394	0.261375	0.232453
Independent U	0.037176	0.131260	0.193594	0.402000	0.191549
Independent R	0.037293	0.098249	0.231350	0.241922	0.391189
Hierarchical U	0.028812	0.111292	0.169270	0.258799	0.121313
Hierarchical R	0.028690	0.076524	0.201372	0.173942	0.240126

Table 6: Ordering violation: absolute bias on the identical 20 datasets. All denominators are 20.

Method	a	b_{A1}	b_{A2}	b_{B1}	b_{B2}
MLE U	0.091063	0.381013	0.052366	0.766011	0.174321
MLE R	0.091063	0.283311	0.716689	0.279834	1.220166
Independent U	0.082042	0.403670	0.058222	0.841169	0.198857
Independent R	0.081104	0.334792	0.796629	0.343048	1.380331
Hierarchical U	0.060655	0.351821	0.078592	0.567210	0.146451
Hierarchical R	0.051623	0.362966	0.764470	0.525702	1.175022

supported primary intervals have 20 valid replications per scenario. Coverage changes in steps of 0.05; at nominal 0.95 its binomial Monte Carlo SE is about 0.049. Narrower intervals cannot be interpreted independently of coverage.

Under the deliberately false restriction, Bayesian credible intervals for the smaller true cause-2 coefficients can have very poor repeated-sampling coverage. An interval is conditional on the assumed restricted model; a nominal 95% posterior probability does not protect against a scientifically false constraint. The exact restricted probability confidence-set intersections can be empty: those are retained as zero coverage and zero set length. All interval families, explicit denominators and negative raw-Wald endpoint flags remain in the supplementary CSV files.

MCMC and computational performance

The historical restricted hierarchical failures were concentrated in order violation, with low bulk ESS and sometimes elevated split rank R-hat. Coordinate-wise truncated Gamma updates near the equality boundary mix more slowly in these cases. Longer fixed-seed runs resolved them without changing the ordering implementation. All monitored model parameters, hyperparameters and reliability transforms in every selected Bayesian fit meet the original criteria. Parameter-wise R-hat, bulk/tail ESS and posterior-mean MCSE are exported; representative traces and posterior densities use original repetition 20 in every scenario.

Runtime totals omit file I/O and orchestration; no precise total wall-clock runtime was captured for the initial completion invocation. The additional 128 fit attempts consumed about 2,371.7 summed process-seconds across the three workers. The actual posterior procedures remain substantially

Table 7: Ordering violation: mean squared error on the identical 20 datasets. All denominators are 20.

Method	a	b_{A1}	b_{A2}	b_{B1}	b_{B2}
MLE U	0.069376	1.000758	0.064089	3.089227	0.348802
MLE R	0.069376	0.396180	0.829559	1.061293	2.471790
Independent U	0.067366	1.069561	0.068385	3.462135	0.394503
Independent R	0.067298	0.421634	1.004535	1.100288	3.120898
Hierarchical U	0.046078	0.811352	0.057145	1.765545	0.179757
Hierarchical R	0.043810	0.350683	0.844107	0.722053	1.924343

Table 8: Strict ordering: CP / AW for marginal 95% intervals. Each available entry has denominator 20; – denotes unsupported.

Method	a	b_{A1}	b_{A2}	b_{B1}	b_{B2}
MLE U	0.95 / 0.872	0.95 / 1.052	0.90 / 2.152	0.95 / 1.835	1.00 / 4.028
MLE R	0.95 / 0.872	–	–	–	–
Independent U	0.95 / 0.852	1.00 / 0.887	0.90 / 1.986	0.95 / 1.533	0.95 / 3.685
Independent R	0.95 / 0.851	1.00 / 0.869	0.90 / 1.978	0.95 / 1.508	0.95 / 3.662
Hierarchical U	0.95 / 0.791	1.00 / 0.874	0.95 / 1.855	0.95 / 1.392	0.95 / 3.214
Hierarchical R	0.95 / 0.798	1.00 / 0.860	0.90 / 1.869	0.95 / 1.375	0.95 / 3.235

Table 9: Equality: CP / AW for marginal 95% intervals. Each available entry has denominator 20; – denotes unsupported.

Method	a	b_{A1}	b_{A2}	b_{B1}	b_{B2}
MLE U	1.00 / 0.834	0.95 / 1.391	0.95 / 1.464	0.95 / 2.415	1.00 / 2.404
MLE R	1.00 / 0.834	–	–	–	–
Independent U	0.95 / 0.816	0.95 / 1.255	0.90 / 1.333	0.90 / 2.172	1.00 / 2.160
Independent R	1.00 / 0.817	0.85 / 0.994	0.95 / 1.368	0.90 / 1.725	1.00 / 2.268
Hierarchical U	1.00 / 0.758	0.95 / 1.207	0.90 / 1.273	0.90 / 1.934	1.00 / 1.948
Hierarchical R	1.00 / 0.766	0.85 / 0.965	0.95 / 1.304	0.85 / 1.559	1.00 / 2.034

Table 10: Ordering violation: CP / AW for marginal 95% intervals. Each available entry has denominator 20; – denotes unsupported.

Method	a	b_{A1}	b_{A2}	b_{B1}	b_{B2}
MLE U	0.85 / 0.864	0.75 / 2.416	1.00 / 1.085	0.80 / 4.491	0.90 / 1.956
MLE R	0.85 / 0.864	–	–	–	–
Independent U	0.90 / 0.851	0.75 / 2.231	0.90 / 0.911	0.90 / 4.085	1.00 / 1.629
Independent R	0.90 / 0.847	0.50 / 1.315	0.35 / 1.458	0.60 / 2.464	0.45 / 2.738
Hierarchical U	0.90 / 0.782	0.80 / 2.046	1.00 / 0.874	0.95 / 3.405	1.00 / 1.404
Hierarchical R	0.90 / 0.778	0.50 / 1.187	0.25 / 1.316	0.55 / 1.977	0.45 / 2.199

Table 11: Observed local fitting cost in seconds. Selected means combine historical and optimized code and different chain lengths; they are not controlled speed benchmarks. All-attempt totals include failures.

Method	Mean selected	Median selected	All-attempt total
MLE U	0.0105	0.0092	0.632
MLE R	0.0099	0.0086	0.594
Independent U	12.6495	10.6131	758.968
Independent R	12.4733	10.6503	748.398
Hierarchical U	75.0175	80.7912	4513.559
Hierarchical R	80.0591	56.9677	6421.928

more expensive than the analytic/profile MLEs.

Interpretation, reproducibility and limitations

The hierarchy has lower observed shape MSE than MLE and the independent prior in all three scenarios, and its unrestricted estimator has lower MSE for every primary parameter than the independent unrestricted estimator in this study. This is evidence for these particular hyperpriors and this small design, not a universal Bayesian advantage. Independent scale-prior Bayesian estimates do not uniformly improve on MLE. Restriction leaves classical shape and totals unchanged; under the independent prior their posterior distributions are also invariant, up to Monte Carlo error from separate method seeds. The hierarchy does not share that invariance.

All primary accuracy comparisons now use identical full-six data-ID sets. No ranking averages errors across parameters with different scales. Twenty replications per scenario, one sample size, and terminal withdrawals only remain major limits. Prior sensitivity, alternate censoring designs and much larger replication counts are needed before broad claims about estimator or interval superiority. Passing diagnostics is not a proof of convergence, and Monte Carlo error remains nonzero.

The saved-result analysis is reproduced locally by:

```
.venv/bin/python -m experiments.completion_study analyze
```

The fitting-resume command, dataset-ID registry, full precision tables and figure index are documented in `docs/SIMULATION_COMPLETION.md`. Historical results are preserved separately.

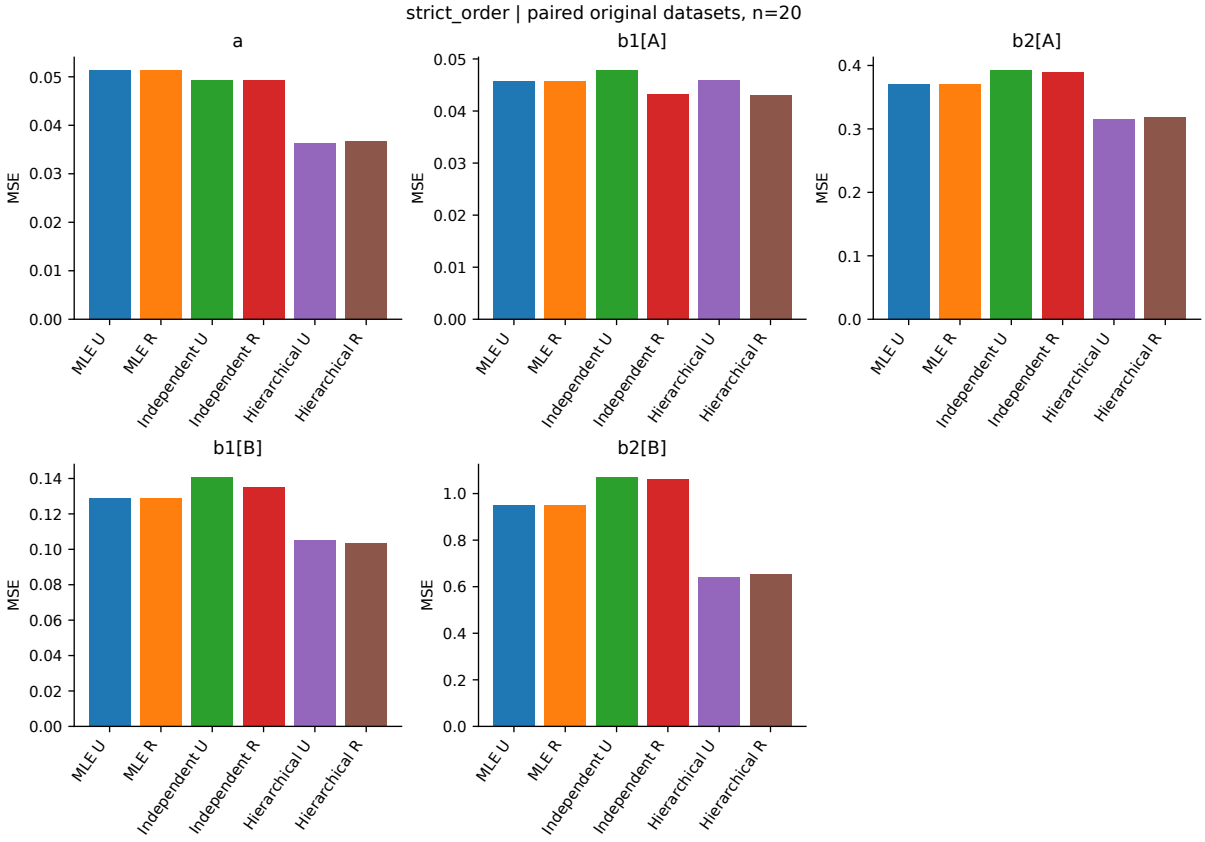


Figure 1: Strict ordering: primary-parameter MSE for all six methods on the same 20 original datasets. U and R denote unrestricted and restricted. Axes start at zero; scales differ by parameter.

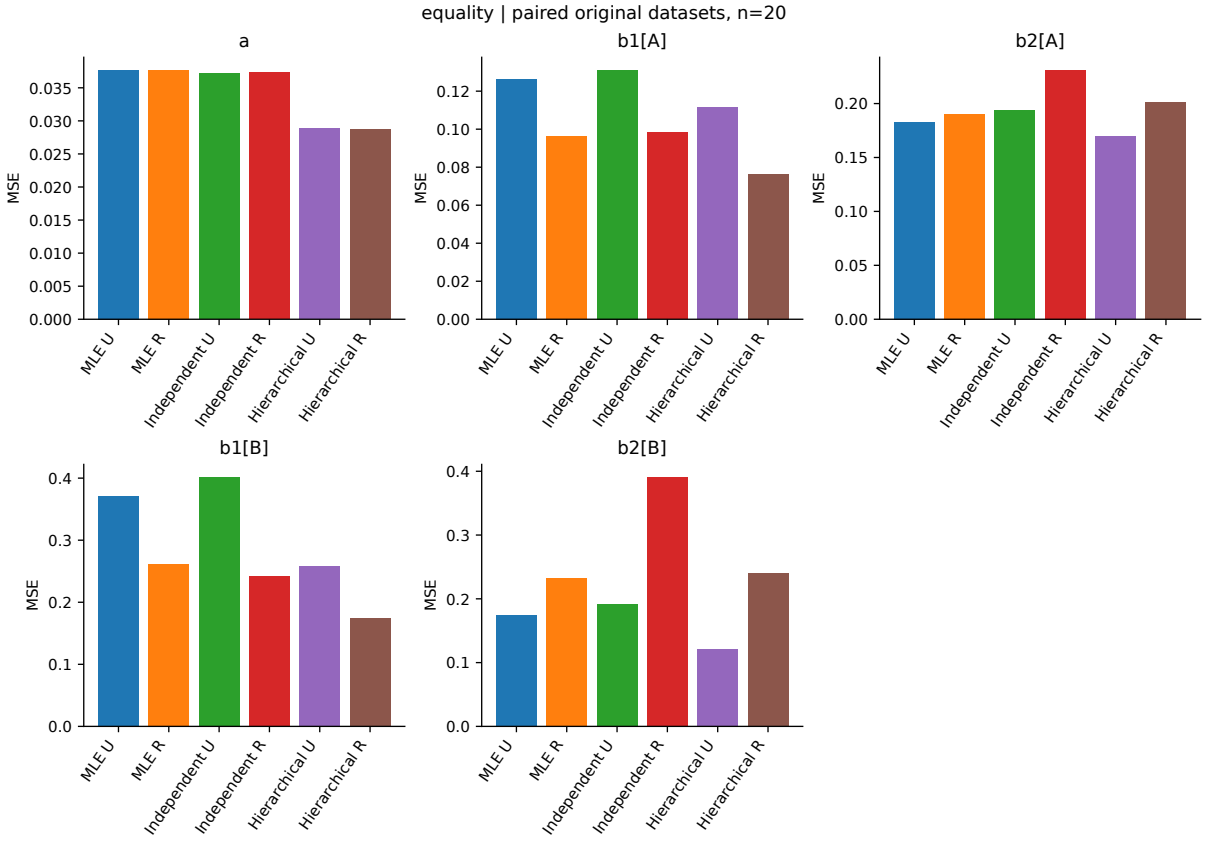


Figure 2: Equality: primary-parameter MSE for all six methods on the same 20 original datasets. U and R denote unrestricted and restricted. Axes start at zero; scales differ by parameter.

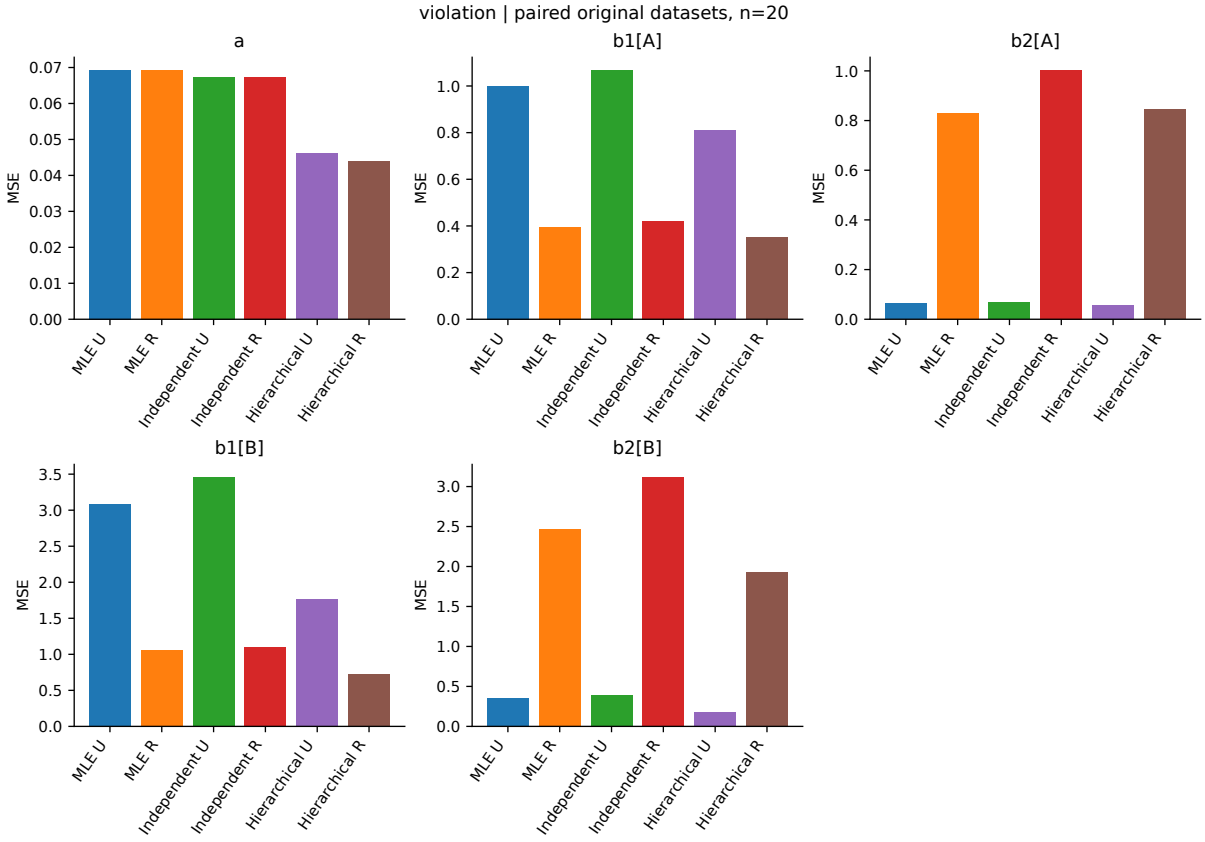


Figure 3: Ordering violation: primary-parameter MSE for all six methods on the same 20 original datasets. U and R denote unrestricted and restricted. Axes start at zero; scales differ by parameter.

5.2 Application to observed appliance failures

We also apply the methods to a published example involving 39 appliance failures. The observations are attributed to Nelson's reliability data and reproduced as Table 6 in a later study by Shi, Yan and Peng [1]. Each observation has a recorded failure time and one of two cause labels: failure mechanism 5 or all other recorded failure mechanisms. There are 26 failures in the first category and 13 in the second.

The source records no withdrawals or unobserved failure causes. We therefore use the *complete-data special case* of our likelihood, with one block, $n = m = 39$ and $r_j = 0$ for all observations. This checks whether the estimation procedures can be applied to genuine failures; it is not a real experiment with progressive removals. We use a single block rather than inventing an additional grouping.

The source reports times in thousands of cycles. For the Kumaraswamy support $(0, 1)$, we divide each value by 5, or equivalently divide the number of cycles by 5,000. The maximum reported lifetime is 4,129 cycles, so the transformed times lie inside the required interval. The 5,000-cycle limit is a chosen analysis scale, *not* a verified physical upper bound. Results should therefore be interpreted relative to this scale.

Table 12: Estimates from the appliance-failure data, verified from the saved computational outputs. Bayesian entries are posterior means.

Method	a	b_1	b_2
Unrestricted MLE	1.0249	1.2567	0.6284
Restricted MLE	1.0249	0.9425	0.9425
Independent Bayes (U)	1.0164	1.2529	0.6244
Independent Bayes (R)	1.0113	0.8835	0.9917
Hierarchical Bayes (U)	1.0340	1.2835	0.6629
Hierarchical Bayes (R)	1.0310	0.9140	1.0262

Table 13: Marginal 95% uncertainty intervals for the appliance data. Classical entries use log-scale approximate confidence intervals; Bayesian entries use shortest contiguous posterior-sample credible intervals. A dash means that the relevant constrained-parameter interval is not supported by the classical procedure.

Method	Interval for a	Interval for b_1	Interval for b_2
MLE (unrestricted)	[0.718, 1.464]	[0.748, 2.110]	[0.330, 1.198]
MLE (restricted)	[0.718, 1.464]	—	—
Independent Bayes (U)	[0.652, 1.376]	[0.667, 1.936]	[0.261, 1.026]
Independent Bayes (R)	[0.651, 1.368]	[0.490, 1.325]	[0.545, 1.474]
Hierarchical Bayes (U)	[0.689, 1.375]	[0.691, 1.931]	[0.293, 1.077]
Hierarchical Bayes (R)	[0.680, 1.384]	[0.494, 1.354]	[0.554, 1.523]

The intervals for a overlap substantially across methods. Under the unrestricted analyses, the intervals for b_1 and b_2 also overlap, so the observed difference in cause frequencies should not be presented as a precise separation between underlying risk parameters. The restriction changes the interpretation of the cause parameters more clearly than it changes the common shape estimate. Importantly, the reported Bayesian intervals summarize posterior uncertainty under each specified prior and model; they are not confidence intervals obtained by inverting repeated sampling tests.

All six estimates were obtained successfully, and the four Bayesian fits passed the recorded diagnostics. The unrestricted classical estimate puts the first cause parameter above the second, matching the larger observed count for mechanism 5. Imposing $b_1 \leq b_2$ forces the classical estimate to $b_1 = b_2$, so it does not fit the observed cause split as closely. The unrestricted log-likelihood is 87.799 compared with 85.590 under the restriction. The two classical fits share the same a and

$B = b_1 + b_2$, hence the same overall fitted survival curve, as predicted by the theory.

The recorded descriptive maximum difference between the fitted and empirical distribution functions is approximately 0.090; this is not a calibrated goodness-of-fit p-value. The source also discusses possible dependence between causes, whereas our latent failure times are assumed independent. Thus this application shows how the methods behave on real observations, but does not prove that the model assumptions hold. We do not compute simulation bias, MSE, or coverage for this dataset because the true parameters are unknown. Table 13 presents the primary marginal uncertainty intervals. Additional reliability predictions and computational diagnostics were recorded during the analysis. The application remains conditional on the selected scaling and independence assumptions.

6 Conclusion

This study provides a common framework for classical and Bayesian estimation with two competing causes and block progressive Type-II censoring. The classical problem becomes simpler after separating total failure risk from the proportions of the two causes. Under the within-block order restriction, the classical total risk and shape estimates stay the same, while cause parameters may be pooled. The independent-scale Bayesian model has a related invariance for overall lifetime quantities, whereas the hierarchical Gamma model does not generally have it.

The appliance example remains unchanged: 26 of 39 failures belong to cause 1, and the stipulated order pools the classical cause coefficients. This restriction requires scientific justification. The completed simulation now has 360 verified valid fits on the original 60 datasets, with all six methods compared on the same 20 datasets per scenario. The hierarchical unrestricted estimator has lower observed MSE than the independent unrestricted estimator for every primary parameter in this design. The order constraint can improve one cause while worsening the other, including at true equality; under false ordering it substantially degrades cause-specific interval coverage. Classical procedures are much cheaper computationally. These descriptive results do not establish a universal best estimator: twenty replications per scenario, one sample size, terminal withdrawals only, and a fixed hierarchical prior remain substantive limitations. Broader designs and prior sensitivity analysis are needed.

References

[1] Shi, Yan and Peng, appliance competing-risks analysis, *Heliyon*, 2025, Table 6. DOI: 10.1016/j.heliyon.2025.e41784. Original observations attributed to W. Nelson, *Applied Life Data Analysis*, 1982.

[2] Wang et al., "Inference for block progressive censored competing risks data from an inverted exponentiated exponential model," *Quality and Reliability Engineering International*, 2023. DOI: 10.1002/qre.3382.

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